

SHAD EBNY WAHID

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RESEARCH PROFILE

Undergraduate electrical engineer working on mixed-signal embedded instrumentation — designing, fabricating and experimentally characterizing hardware where analog front-end behavior, power stages and real-time firmware must be co-designed. Built a complete low-cost ultrasonic phased-array imaging platform from schematic to image formation. Research interests: mixed-signal and power electronics hardware, embedded sensing instrumentation, and array signal processing.

EDUCATION

B.Sc. in Electrical and Electronic Engineering

Jamalpur Science and Technology University (formerly Bangamata Sheikh Fojilatunnesa Mujib Science and Technology University), Jamalpur, Bangladesh

Session 2022–2023 · Batch EEE04 · Expected graduation: 2028

CGPA: 3.2 / 4.00 through 2nd year, 2nd semester

Relevant coursework: Power Electronics, Electrical Machines, Control Systems, Digital Signal Processing, Communication Engineering, Electromagnetics, Digital Electronics, Power Systems

PUBLICATIONS

Under review

1. N. F. Iva, R. Khan, A. Aman, A. Ananto, **S. E. Wahid**, and M. M. Haque, “A unified assistive platform for visually impaired individuals: integrating autonomous navigation, currency recognition, and real-time GPS tracking,” submitted to 29th International Conference on Computer and Information Technology, 2026. (*under review*)

In preparation

2. **S. E. Wahid**, J. R. G. Bristy, and M. M. Haque, “Per-element acoustic calibration of a 40 kHz ultrasonic transmit array,” targeted at *IEEE Sensors Letters*, 2027.

RESEARCH EXPERIENCE

Low-Cost Ultrasonic Phased-Array and 2D Acoustic Imaging Platform — 2025–present

Undergraduate research; manuscripts in preparation

- Designed and experimentally evaluated an air-coupled 40 kHz ultrasonic imaging platform using an 8-transmitter / 1receiver architecture with ESP32-class embedded acquisition.
- Designed and built the complete transmit and receive electronics: MOSFET-driven transmit channels and TL072-based analog receive conditioning, on a custom mixed-signal PCB physically split to suppress acoustic and electrical crosstalk, with star-ground topology.
- Diagnosed and corrected a DC bias fault in the receive envelope-detector chain through series-capacitor isolation and divider-resistor selection, restoring receive dynamic range.
- Implemented cycle-accurate transmit beamforming under FreeRTOS with multi-mode aperture control (full-8, left-4, right-4 sub-apertures) and an MCP3008 SPI acquisition chain.
- Collected and analysed multi-channel waveform datasets: phase and amplitude consistency, reflector response, background envelopes, angular sweeps and array-diversity behaviour; characterised a systematic time-of-flight offset and identified a firmware-side correction.
- Investigated target localisation, reflector discrimination, grating-lobe suppression and 2D image formation under severe spatial-sampling and low-cost hardware constraints.
- Built reproducible Python acquisition and processing workflows for raw waveform capture, metadata handling, feature extraction, quality control and plotting.

Real-Time Sonar Signal Processing Pipeline — 2025

- Developed a GPU shader-based real-time sonar display implementing delay-and-sum beamforming, moving-targetindication clutter cancellation, CA-CFAR detection and Kalman-filter track maintenance.
- Extended to a fan-sector 2D acoustic imager with heatmap rendering and interactive gain control.

First-Principles DFT Study of Cs_3MX_3 ($\text{M} = \text{Au, Pt, Pd}$) Under Tensile Strain — 2025–2026

Quantum ESPRESSO, under Md. Mahfuzul Haque

– Performed SCF and band-structure calculations for Cs_3AuX_3 , Cs_3PtX_3 and Cs_3PdX_3 across a range of isotropic tensile strain states; built and managed the multi-strain-point convergence workflow.

ENGINEERING PROJECTS

EI-Core Transformer — Design, Winding and Experimental Characterisation — 2025

Designed and hand-wound an EI-core transformer to specification; performed open-circuit and short-circuit tests to extract equivalent-circuit parameters and evaluate practical loss and regulation behaviour.

Bangladeshi Currency Recognition System — 2026

Machine-learning classifier for Taka banknote detection and denomination classification.

nat_audit.py — ISP Network Transparency Audit Tool — 2026

Python tool auditing NAT/CGNAT deployment and BDIX peering behaviour across Bangladeshi ISPs, producing a comparative report.

PROFESSIONAL EXPERIENCE

PCB Designer (paid), Early Stroke-Risk Detection Wristband — Research team · 2026–present

Designed the 4-layer wristband PCB in KiCad, integrating an ESP32-S3 module, MAX30102 PPG sensor, single-lead ECG front-end and MPU6050 IMU with OLED display, USB-C charging and LiPo power; delivered a DRC-clean, manufacturing-ready package for the team's planned patient study.

Freelance PCB Designer — Fiverr · 2026–present

Schematic capture and multi-layer PCB layout for external clients.

Videographer / Production Assistant — JSTU · 2025–2026

Produced official departmental and university documentary films on two occasions; the second was a paid commission.

HONOURS & AWARDS

2nd Place, Single-Axis Solar Tracking Panel, Department of EEE project competition, JSTU

Awarded at the university's training programme on net metering for on-grid solar systems.

Documentary Making Award, Jamalpur Science and Technology University

Recognised for the documentary films produced for the department and university.

LEADERSHIP

Joint General Secretary, Green Voice–JSTU — 2025–2026

Organisational communication, event coordination and collaborative programme delivery.

Assistant Event Manager, JSTU Robotics Club — 2025

Logistics, scheduling and volunteer coordination for departmental and university robotics events.

Class Representative, Department of EEE — 2023

Elected representative for one academic year. Project group leader on two multi-member design projects; repeated laboratory group leader.

TECHNICAL SKILLS

Hardware: Schematic capture, multi-layer PCB layout, mixed-signal design, grounding and EMI practice, analog front ends, op-amp circuits, MOSFET switching, magnetics design and characterisation, SMD and through-hole soldering, board bring-up and fault diagnosis

Embedded: ESP32 / ESP32-S3, ESP8266, Arduino, FreeRTOS, SPI / I²C / UART, ADC acquisition chains, real-time timing control

Signal Processing: Delay-and-sum beamforming, envelope detection, CA-CFAR, Kalman filtering, time-of-flight estimation, waveform and experimental data analysis

Simulation & Computing: LTspice, MATLAB/Simulink, Proteus, Quantum ESPRESSO (DFT), Python scientific stack

Software: Python, C/C++, Kotlin (Jetpack Compose), Processing, LaTeX, Git, Linux

Languages: Bangla (native), English — IELTS scheduled June 2028, target 7.0

REFEREES

Md Mahfuzul Haque, Assistant Professor
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